Compactness through finite intersections

The complete blueprint

All levels · read along the `refines` order · appendices included

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Contents

1 Introduction 1

2 Covers and compact spaces 1

2.1 Covers . . . 1

2.2 Compact spaces . . . 2

2.2.2 Compactness by closed sets . . . 2

3 Products 3

4 Conventions 3

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1 Introduction

We prove that a space is compact exactly when every family of closed sets with the finite intersection property has a point in common (Theorem 2.2.2), and use it to show that a product of two compact spaces is compact (Theorem 3.1). The argument is the one in Čech's and Erdős' lecture notes, written so that every step can be checked on its own.

Throughout, X and Y are topological spaces and $\mathcal{O}(X)$ is the set of open subsets of X.

2 Covers and compact spaces

The vocabulary of covers, and the characterisation of compactness by closed sets.

2.1 Covers

Two notions recur below, and both are about families of subsets indexed by an arbitrary set I.

Definition 2.1.1 (Open cover)**.** An *open cover* of X is a family $\mathcal{U} = (U_i)_{i\in I}$ of open sets with

$$\bigcup_{i\in I} U_i = X.$$

A *subcover* is a subfamily $(U_j)_{j\in J}$, $J \subseteq I$, that is still a cover; it is *finite* when J is.

Definition 2.1.2 (Finite intersection property)**.** A family $(F_i)_{i\in I}$ of subsets of X has the *finite intersection property* when every finite subfamily meets:

$$F_{i_1} \cap \cdots \cap F_{i_n} \neq \emptyset \quad \text{for all } n \geq 1 \text{ and } i_1, \ldots, i_n \in I.$$

It is the dual of Definition 2.1.1 under complements.

2.2 Compact spaces

Definition 2.2.1 (Compact space)**.** X is *compact* when every open cover of X (Definition 2.1.1) has a finite subcover.

Passing to complements turns open covers into families of closed sets, and "has no finite subcover" into "has the finite intersection property". The next theorem records this.

Theorem 2.2.2 (Compactness by closed sets)**.** *X is compact if and only if every family $(F_i)_{i\in I}$ of closed subsets of X with the finite intersection property has $\bigcap_{i\in I} F_i \neq \emptyset$.*

Proof 2.2.2.1. Of Compactness by closed sets (Theorem 2.2.2).

Both directions are the contrapositive of each other once the complements are taken: Lemma 2.2.2.1.1 exchanges the two conditions, and Lemma 2.2.2.1.2 says what a finite subcover becomes.

USES → Finite intersection property (Definition 2.1.2)
Where the finite intersection property enters. Only in Lemma 2.2.2.1.2: the property is the statement that no finite subfamily has empty intersection, word for word.

Lemma 2.2.2.1.1 (Complements)**.** *For a family $(F_i)_{i\in I}$ of closed sets put $U_i = X \setminus F_i$. Then*

$$\bigcap_{i\in I} F_i = \emptyset \iff \bigcup_{i\in I} U_i = X,$$

so (U_i) is an open cover exactly when the F_i have no common point.

Lemma 2.2.2.1.2 (Finite subfamilies)**.** *For $J \subseteq I$ finite, $(U_j)_{j\in J}$ covers X if and only if $\bigcap_{j\in J} F_j = \emptyset$. Hence (U_i) has no finite subcover if and only if (F_i) has the finite intersection property.*

■

Lemma 2.2.3 (Tube lemma)**.** *Let Y be compact, $x \in X$, and $W \subseteq X \times Y$ open with $\{x\} \times Y \subseteq W$. Then there is an open $U \ni x$ with $U \times Y \subseteq W$.*

Proof. Cover $\{x\} \times Y$ by basic open sets $U_y \times V_y \subseteq W$. The V_y cover Y, so finitely many do, say $V_{y_1}, \ldots, V_{y_n}$, and $U = U_{y_1} \cap \cdots \cap U_{y_n}$ is the open set wanted. ■

3 Products

Finite products of compact spaces are compact. The tube lemma (Lemma 2.2.3) is stated in Section 2.2; it is the whole content of the argument.

Theorem 3.1 (Products of two compact spaces)**.** *If X and Y are compact, so is $X \times Y$ with the product topology.*

Proof. Let $\mathcal{W}$ be an open cover of $X \times Y$. For each $x \in X$ the slice $\{x\} \times Y \cong Y$ is compact, so finitely many members of $\mathcal{W}$ cover it; call their union W_x. By Lemma 2.2.3 there is an open $U_x \ni x$ with $U_x \times Y \subseteq W_x$.

1. The U_x cover X, so by Definition 2.2.1 finitely many do: $U_{x_1}, \dots, U_{x_m}$.
2. Then $X \times Y = \bigcup_{k=1}^{m} U_{x_k} \times Y \subseteq \bigcup_k W_{x_k}$, a union of finitely many members of $\mathcal{W}$.

Compare the closed-set form, Theorem 2.2.2, which gives the same result by a less direct route. ∎

Remark 3.2 (Arbitrary products)**.** For an arbitrary family $(X_\alpha)_{\alpha \in A}$ the statement is Tychonoff's theorem, which needs the axiom of choice; the argument above only gives finite products, by induction on the number of factors: $\prod_{k=1}^{n+1} X_k \cong (\prod_{k=1}^{n} X_k) \times X_{n+1}$.

Lemma 2.2.3 (Tube lemma) belongs here too; it is written out under 2.2 Compact spaces, on page 2.

4 Conventions

Notation used above, collected in one place.

Remark 4.1 (Symbols)**.**

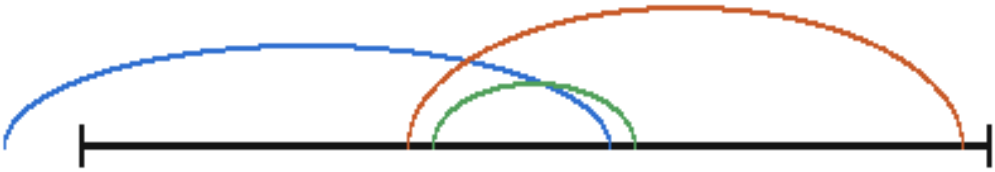
Symbol	Meaning	First used
$\overline{A}$	the closure of A	Definition 2.1.2
$\mathcal{O}(X)$	the open subsets of X	Section 1
$\mathcal{U}$, $\mathcal{W}$	families of open sets	Definition 2.1.1
Top	the category of spaces, $f\colon X \to Y$ continuous	—

Greek letters index families ($\alpha \in$ A), and $\mathbb{R}$, $\leq$, $\to$ are used as usual. Identifiers in the Lean sources look like `TopologicalSpace.IsCompact` or `isCompact_iff_finite_subfamily_closed`; a typical lookup is

```
example (X : Type*) [TopologicalSpace X] :
    CompactSpace X ↔ IsCompact (Set.univ : Set X) := ⟨fun h => h.1, fun h => ↩
  ⟨h⟩⟩
```

Every result here is classical; see any text on point-set topology.

Remark 4.2 (A finite subcover)**.** The picture shows three open sets covering a compact interval; two of them already suffice.



Three open intervals covering a closed interval